

B.Tech. Degree II Semester Regular Examination July 2**EE/EC/CS/IT 23-200-0205B INTRODUCTION TO CYBER PHYSICAL SYSTEM
(2023 Scheme)**

Time: 3 Hours

Maximum M

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand the features and components of Cyber Physical Systems.
 CO2: Understand the elementary constructs of Arduino Software.
 CO3: Develop optimal programs and circuits for interfacing various sensors with Arduino.
 CO4: Apply Arduino IDE for developing suitable programs for interfacing sensors and actuators with Arduino and Node MCU.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

(5 × 2 = 10) Marks BL

- | | | | |
|--------|---|---|----|
| I. (a) | Compare active and passive sensors in IoT using a suitable example. | 2 | L1 |
| (b) | Describe the general structure of an Arduino program. | 2 | L1 |
| (c) | Contrast the analogRead() function with digitalWrite() in Arduino. | 2 | L1 |
| (d) | Write a NodeMCU code to blink an LED at 1-second intervals. | 2 | L1 |
| (e) | How do you set up and use the Arduino IDE for programming Arduino boards? | 2 | L1 |

PART B

(4 × 10 = 40)

- | | | | |
|---------|--|---|----|
| II. (a) | Explain the features of Cyber Physical System. | 6 | L1 |
| (b) | What are the key characteristics that define the Internet of Things (IoT)? | 4 | L1 |

OR

- | | | | |
|----------|---|---|----|
| III. (a) | Define Cyber-Physical Systems (CPS). Explain the general structure of CPS. | 6 | L1 |
| (b) | List and describe some of the key enabling technologies that support IoT implementations. | 4 | L1 |

- | | | | |
|---------|---|---|--|
| IV. (a) | Compare and contrast the applications of delay(), millis(), micros() and delayMicroseconds() in Arduino projects. | 4 | |
| (b) | Explain the various types of operators used in Arduino programming. | 6 | |

OR

- | | | | |
|--------|--|---|--|
| V. (a) | List and explain the basic data types supported by Arduino. | 5 | |
| (b) | Explain the functionality of these components on an Arduino board: | 5 | |
| (i) | Crystal oscillator. | | |
| (ii) | Voltage regulator. | | |
| (iii) | TX RX LEDs. | | |
| (iv) | AREF. | | |

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		Marks
VI.	(a) Describe how PWM works on Arduino and its applications.	5
	(b) How do you integrate an ultrasonic sensor with Arduino for distance measurement?	5
OR		
VII.	(a) Discuss the process of interfacing humidity and temperature sensors with Arduino.	6
	(b) What are pull-up and pull-down resistors and how do they affect push button switch interfacing?	4
VIII.	(a) List the main differences between Arduino Uno and Node MCU development boards.	5
	(b) Explain how to connect NodeMCU to a WiFi network.	5
OR		
IX.	(a) Write an Arduino code to control a servo motor.	5
	(b) How can you interface a gas sensor with NodeMCU?	5

Bloom's Taxonomy Levels

L1 – 14%, L2 – 43%, L3 – 19%, L4 – 14%, L6 – 10%.
